COGNITIVE SYSTEMS ENGINEERING • MONOGRAPH

Extrinsic Education vs. Intrinsic Rationality in Autonomous AI

Why Frozen Language Models Need Cognitive Architecture: Reconciling "Diagnose-Then-Teach" with Brain-Inspired Governance

Zhao Wan • Dept. of Computer Science | **Cosmos Analytics AI Lab** • Cognitive Systems Group

Date: November 2026

THE CORE THESIS: YOU CANNOT PROMPT AN AI INTO BEING RATIONAL

`EDUCATION = CONTENT • RATIONALITY = GOVERNANCE`

1. EDUCATION IS EXTRINSIC

`The Curriculum`

Education is **what is poured into the model from the outside**: system prompts, rulebooks, handbooks, SOP guidelines, and precedent cases. It is the external knowledge payload.

Key Trait: Passive ingestion. An LLM can memorize all 16 rules in a handbook and still be gullible, reckless, and blind to contradictory local facts.

2. RATIONALITY IS INTRINSIC

`The Governance`

Rationality is **how the mind governs itself from the inside**: self-doubt, calibration, measuring rollout conflict, and balancing the asymmetric cost of mistakes before acting.

Key Trait: Active executive control. A rational agent defaults to **HOLD**, verifies whether confidence matches reality, and escalates to humans when uncertain.

The "Educated Fool" Trap: An AI crammed with extrinsic rules without intrinsic rationality will blindly obey written text over code reality, hallucinate high confidence, and commit disastrous mistakes.

1. THE EMPIRICAL EVIDENCE: WHAT HAPPENS WHEN EDUCATION LACKS RATIONALITY?

Zhao Wan (October 2026)

In a rigorous empirical evaluation of 1,848 code repair attempts on Claude Sonnet and Gemini 2.5 Flash across 14 bug families, Wan (2026) demonstrated both the power and the severe failure modes of purely extrinsic instruction:

1 **Gain Tracks Headroom**

`Gain ≤ 100 − Baseline`

Where models knew nothing unaided (0% baseline), instruction produced massive gains (+60 to +100 points). But where the model already scored 100%, instruction yielded zero gain.

2 **Pathological Over-Compliance**

`100% → 56% (-44 points)`

In `service_errors`, Claude scored 16/16 unaided, but crashed to 9/16 with its rule. The model blindly obeyed the extrinsic prompt rule over empirical ticket evidence!

3 **Handbook Cross-Interference**

`All 16 Rules Broke Code`

Dumping all 16 rules scored worse than 1 rule in 4 families. Rules leaked unneeded error imports, container-immutability leaked into state machines, and placebos cost Gemini points ($p \approx 0.04$).

2. THE INTRINSIC SOLUTION: THE 3 COGNITIVE HARNESSES

Brain-Inspired Governance

To transform an educated model into a truly rational agent, the Cognitive Architecture bounds the frozen foundation LLM inside three autonomous harnesses:

1 KNOW IT (Education)

Ingestion

Cortex (Semantic Memory)
Ontology + Knowledge Graph (Invariant domain rules & SOPs)

Hippocampus (Episodic Memory)
Context Graph index of past cases; *opened only when needed*

Fix: Stops context stuffing by retrieving only matching invariants.

2 BE SURE (Calibration)

Epistemic

Prefrontal Metacognition
Reliability curve: says 90% only when right ~90% of the time

Anterior Cingulate Monitor
Samples N=5 rollouts. If answers disagree, knowledge has not taken hold

Fix: Stops over-compliance by evaluating confidence vs reality.

3 ACT (Gate Harness)

Instrumental

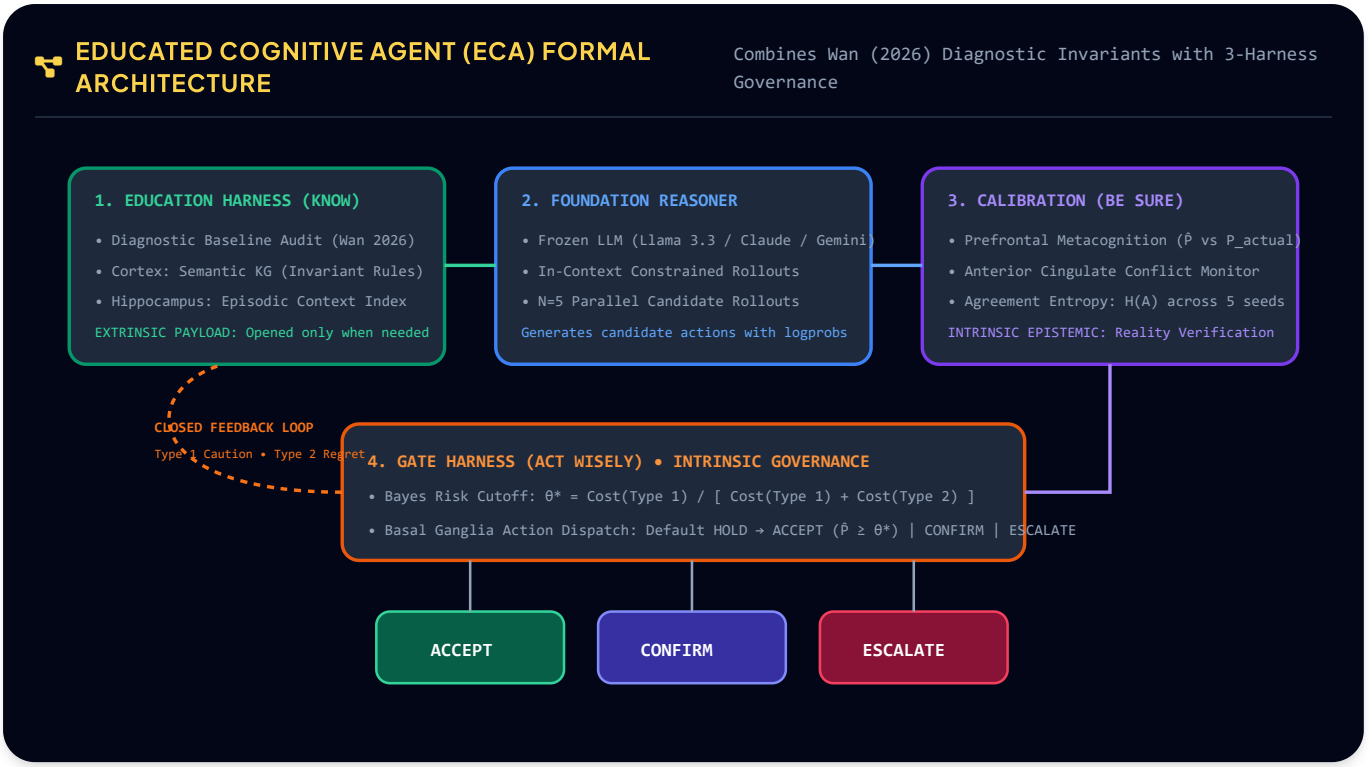
Risk Threshold (θ*)
$\theta^* = \text{Cost(Type 1)} / [\ \text{Cost(Type 1)} + \text{Cost(Type 2)}\ ]$

Basal Ganglia Dispatch
Default is **HOLD**. Releases to ACCEPT only when $\hat{P} \geq \theta^*$, else ESCALATE

Fix: Stops un-gated action on unsolvable cases (e.g. task 2018).

3. THE SYNTHESIS: THE EDUCATED COGNITIVE AGENT (ECA)

Extrinsic Invariants Governed by Intrinsic Reason



4. LEARNING FROM MISTAKES: HOW WEAK SPOTS GO BACK TO KNOW

Reconciling "Growing Backward to Grow Forward"

In production, an agent will inevitably make mistakes. The Cognitive Architecture implements a dual dopaminergic feedback loop that mirrors human adult learning:

Bad Item Accepted (Type 1 Error)

Caution: Raise the Bar

When the agent approves a bad patch or fraudulent claim, it activates **Caution**:

- Cost of Type 1 is increased → pushes threshold θ* higher.
- A defensive invariant rule is automatically injected into the Cortex KG.

Good Item Held Back (Type 2 Error)

Regret: Lower the Bar

When held-back items are checked against the ground truth answer key and found valid:

- Agent experiences **Regret**: lowers θ* to prevent future timidness.
- The missed edge case is indexed directly into the Hippocampus Context Graph.

5. COMPARATIVE EVALUATION: EXTRINSIC EDUCATION VS. COMBINED RATIONALITY

Capability Dimension	Pure Extrinsic Education (Wan 2026)	Combined Architecture (Educated Cognitive Agent)
Knowledge Delivery	Static handbook string injected into prompt	**Dynamic dual-graph grounding (Cortex rules + Hippocampus precedents)**
Interference Defense	Vulnerable to full-handbook leakage & placebo clutter	**Selective precedent indexing opens rules only when needed**
Over-Compliance Defense	None (Claude crashed -44 pts on service_errors)	**Prefrontal Metacognition discounts rules that contradict local facts**
Sampling Variance	Single-shot attempts subject to stochastic noise	**Anterior Cingulate checks agreement entropy across N=5 rollouts**
Action Selection	Un-gated (all generated patches deployed)	**Bayes risk threshold (θ*) with Basal Ganglia default HOLD**
Unsolvable Problems	Repeated failure (Task 2018 failed 0/8 across all arms)	**Automatic ESCALATE to human specialists when $\hat{P} < \theta^*$**
Mistake Correction	Offline manual handbook authoring	**Real-time Closed Feedback Loop (Caution & Regret parameter tuning)**

6. THREE PRACTICAL GOLDEN RULES FOR AI ENGINEERS

> **RULE 1:** **Do not prompt when headroom is zero.** If a model already scores 95%+ on a task unaided, injecting prompt rules provides zero upside and invites cross-rule regression. Run a diagnostic baseline first.

> **RULE 2:** **Never trust self-reported confidence.** Measure multi-rollout consensus entropy (N=5). If rollouts disagree, the knowledge has not taken hold, regardless of how confident the tokens look.

> **RULE 3:** **Rationality is architectural, not verbal.** You cannot prompt an agent into being careful by saying "please double check." Risk gating must be enforced by a structural Bayes gate that defaults to HOLD.

REFERENCES

1. Wan, Z. (2026). *Instruction Gains Track the Knowledge Gap: A Diagnose-Then-Teach Analysis of Frozen Language Models.* Research Report, October 2026.

2. Cosmos Analytics AI Laboratory. (2026). *A Rational AI Agent Works Like a Brain: Cognitive Architecture Whitepaper.*

3. Kahneman, D. (2011). *Thinking, Fast and Slow.* Farrar, Straus and Giroux.

4. Brier, G. W. (1950). Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1), 1-3.